

TCS3351 CRYPTOGRAPHY AND DATA SECURITY
Tutorial 06
SYMMETRIC BLOCK CIPHER - MODES OF OPERATION

1. Explain why modes of operation are needed if modern block ciphers are used for encryption.
2. Briefly explain Electronic Code Book (ECB) mode, list its advantages and disadvantages, and prove that the plaintext used by Alice can be recovered by Bob at the receiving end.
3. Consider using the ciphertext stealing technique for Electronic Code Book mode.
 - (i) Show the diagram for encryption and decryption for the last two blocks.
 - (ii) Show the equations used for encryption and decryption algorithm for the last two blocks.
4. (i) Briefly explain Cipher Block Chaining (CBC) mode, list its advantages and disadvantages, and prove that the plaintext used by Alice can be recovered by Bob at the receiving end.
(ii) Assume that bit 5 in ciphertext block 3 is corrupted during transmission. Identify the plaintext blocks affected by this.
5. Consider using the ciphertext stealing technique for Cipher Block Chaining (CBC) mode.
 - (i) Show the diagram for encryption and decryption for the last two blocks.
 - (ii) Show the equations used for encryption and decryption algorithm for the last two blocks.
6. Differentiate between Cipher Feedback (CFB), Output Feedback mode (OFB), and Counter (CTR) modes of operation.
7. Divide the five modes of operation (ECB, CBC, CFB, OFB, CTR) into two groups based on the following:
 - a. Those that use the encryption and decryption functions of the underlying cipher and those that use only the encryption function.
 - b. Those that need padding and those that do not
8. Draw the diagram for encryption and decryption for the following modes of operation.
 - (i) Cipher Feedback Mode if the size of plaintext block (r) is equal to the size of the block used in block cipher (n)
 - (ii) Output Feedback Mode if the size of plaintext block (r) is equal to the size of the block used in block cipher (n)

References:

1. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
2. William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017